A TERMWISE SPLIT SEQUENCE OF COMPLEXES OF COMPLEXES GIVES A CONNECTING MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; both identities it uses are displayed inside the counted span and the closing words 'as desired' follow the computation that establishes the claim, so no step is deferred.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Editorial note. Same-name and twin check: the chapter also states lemma-ses-termwise-split at source lines 3578-3598, which constructs the same connecting morphism in the homological grading (a morphism C to A[-1]). The two statements have the same conclusion under the reindexing A_n = A^{-n} and equivalent splitting hypotheses, so per the lane rule for twins only one of them is counted, namely this cohomological version. The homological twin is recorded in lane-s4/DISPOSITIONS-stacks-homology.json and is not counted anywhere.

Allowed standard prerequisites (editorial).

- Splittings of a complex of complexes. Same chapter, the discussion of a complex of complexes and of compatible direct sum splittings of its terms. Supplies the maps s^n and pi^n induced by the splitting B^n = A^n plus C^n that the construction uses.
- Differentials and the shift of a complex. Same chapter, the definition of the differential of a complex and of the shift A[1]. Supplies the objects C^n and A[1]^{n+1} connected by the constructed family and the sign convention used in the computation.

Lemma 0.1. *Let $\mathcal{A}$ be an additive category. Let*

$$0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$$

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be a complex (!) of complexes. Suppose that we are given splittings $B^n = A^n \oplus C^n$ compatible with the maps in the displayed sequence. Let $s^n : C^n \to B^n$ and $\pi^n : B^n \to A^n$ be the corresponding maps. Then the family of morphisms

$$\pi^{n+1} \circ d_B^n \circ s^n : C^n \to A^{n+1}$$

define a morphism of complexes $\delta : C^\bullet \to A[1]^\bullet$.

Proof. Denote $i : A^\bullet \to B^\bullet$ and $q : B^\bullet \to C^\bullet$ the maps of complexes in the short exact sequence. Then $i^{n+1} \circ \pi^{n+1} \circ d_B^n \circ s^n = d_B^n \circ s^n - s^{n+1} \circ d_C^n$. Hence $i^{n+2} \circ d_A^{n+1} \circ \pi^{n+1} \circ d_B^n \circ s^n = d_B^{n+1} \circ (d_B^n \circ s^n - s^{n+1} \circ d_C^n) = -d_B^{n+1} \circ s^{n+1} \circ d_C^n$ as desired. □